THE HONG KONG POLYTECHNIC UNIVERSITY
Department of Electronic and Information Engineering

EIE4121 Machine Learning in Cyber-security

Mini-project

The mini-project is a critical component of this course. The work should reflect students' understanding of specific topics beyond the level/scope covered in lectures. The project offers a hands-on opportunity for students to apply their knowledge and showcase their problem-solving skills in the realm of machine learning and data analysis.

Details

Large language models (LLMs) have been utilized in various applications, including generating emails and texts. While LLMs can be very useful, they have been used to create phishing emails. In this project, the aim is to define a classification problem involving LLM-generated text and determine whether machine learning methods can be used to detect whether an LLM has generated the text.

The dataset can be downloaded at

https://www.kaggle.com/datasets/francescogreco97/human-llm-generated-phishing-legitimate-emails/data.

This dataset includes a multiclass email dataset of

- 4000 Human-generated and
- LLM-generated emails

of both phishing and genuine nature. The Human-generated emails were obtained by selecting the most recent emails from the Nazario and Nigerian Fraud datasets (1,000 legitimate and 1,000 phishing).
The LLM-generated emails were generated by ChatGPT and WormGPT (1000 legitimate + 1000 phishing).

You can read this reference to understand how machine learning methods can be used to study the dataset: "David versus Goliath: Can Machine Learning Detect LLM-Generated Text? A Case Study in the Detection of Phishing Emails" (2024) F. Greco, G. Desolda, A. Esposito, A. Carelli, ITASEC 2024 conference on cybersecurity.

Two students will form a group. There are two phrases in the mini project.

<u>Phase 1: Project proposal (due date: 13 March 2026)</u>

Each group will prepare a project proposal, consisting of the following main items:

- An outline of the classification problem to be addressed.

- Perform a feature analysis on the data and investigate whether there are differences among the classes.
- Identify two methods that would be used to solve the classification problem.

Example classification problems to be solved are:

- Distinguish whether the email is generated by a human or LLMs
- Distinguish whether the human-generated email is phishing or not.
- Distinguish whether the LLM-generated email is phishing or not.

The proposal is approximately three pages in length.

Phase 2: Project Report and Presentation (due date: 10 April 2026)

A group should submit one project report, one video (approximately 8-10 minutes in length), and Python code to the Blackboard.

(1) Python notebook file (***.ipynb) containing all results (with comments)
(2) A short report summarizing the two methods employed to solve the problem (around two pages), and comparing their results (2 to 3 pages). You would also need to identify the limitations of the two methods. The whole report should be less than 10 pages.
(3) A 10-minute video presentation detailing the methods used, the results obtained, and a discussion of the results. Note that each student should give a verbal presentation, and their identity should be clear in the video. The presentation will be assessed individually.
(4) A contribution table in the following table format:

Name of the student	works performed	Contribution to the whole project (in terms of percentage)

Marks distribution

- Proposal: 2%
- Report: 12%
- Presentation: 5%

  Total: 19%